

C1 Four Topology MOSbius

Schematic Review Slides 07/03

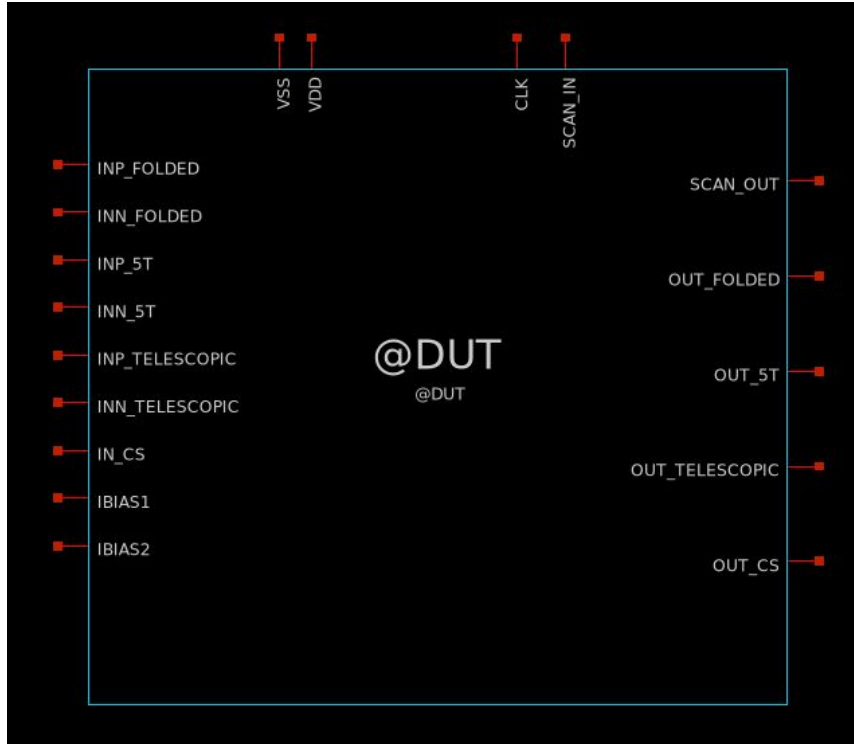
[LINK to Video](#)

Discord name	Affiliation	Experience	Role
elij0100	Columbia University	Undergraduate	Team Lead
boibun.	Columbia University	Undergraduate	Team Member
maxdrucker_83950	Columbia University	Undergraduate	Team Member
Manwell	Columbia University	Undergraduate	Team Member

Project Information

Goal: To design and fabricate a chip containing several of the most common amplifier topologies, all with accessible pins, so that a student can wire up, test, characterize, and compare each amplifier. Write a short lab manual to accompany the chip (consider during design, but can follow design/layout submission). Students measure DC gain, GBW, slew rate, stability, PSRR, etc. for each amplifier, learning the trade-offs through hands-on experimentation

System Overview



Total Pins: 17

Global Input (2): VDD, VSS

Digital Input (2): Scan In, Clock

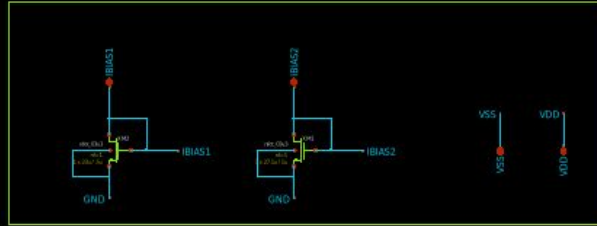
Digital Output (1): Scan Out

Analog Input (8): IREF, IN(CS), INP/INN (5T), INP/INN (Telescopic), INP/INN (Folded)

Analog Output (4): OUT (CS), OUT (5T), OUT (Telescopic), OUT (Folded)

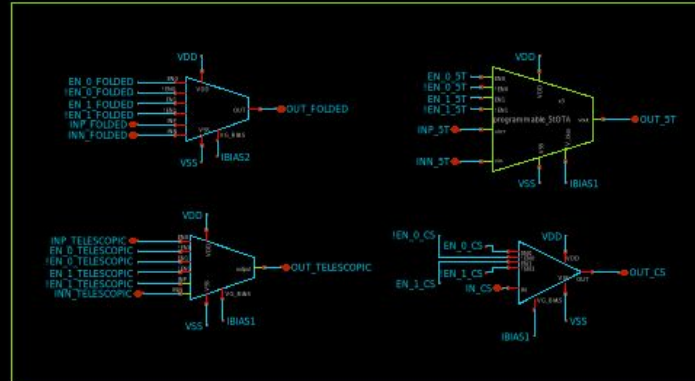
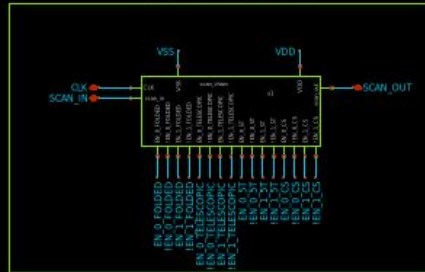
System Block Diagram

BIAS AND POWER PINS



CORE AMPLIFIERS

SCAN CHAIN

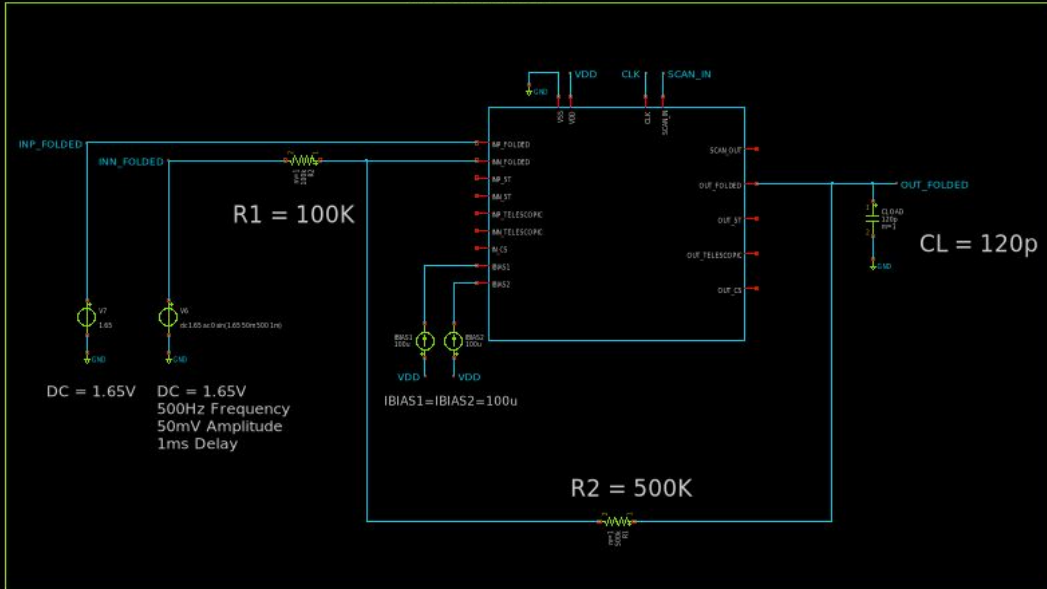


In practice, we will not have two separate bias currents. At the moment, there is some mismatch between the initial current mirrors in the amplifiers. This will be fixed so that we use only one externally generated bias current for the entire chip.

System Testbench

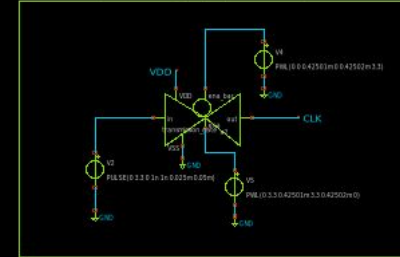
IMPLEMENTATION

Configure the folded cascode in 3x mode as an inverting amplifier to amplify a sine wave input by 5 while driving a 120pF capacitive load.



CLOCK GENERATION

Artificially generate a clock signal and ensure it turns off after around .425ms, which is when the scan in signal has fully propagated through the scan chain

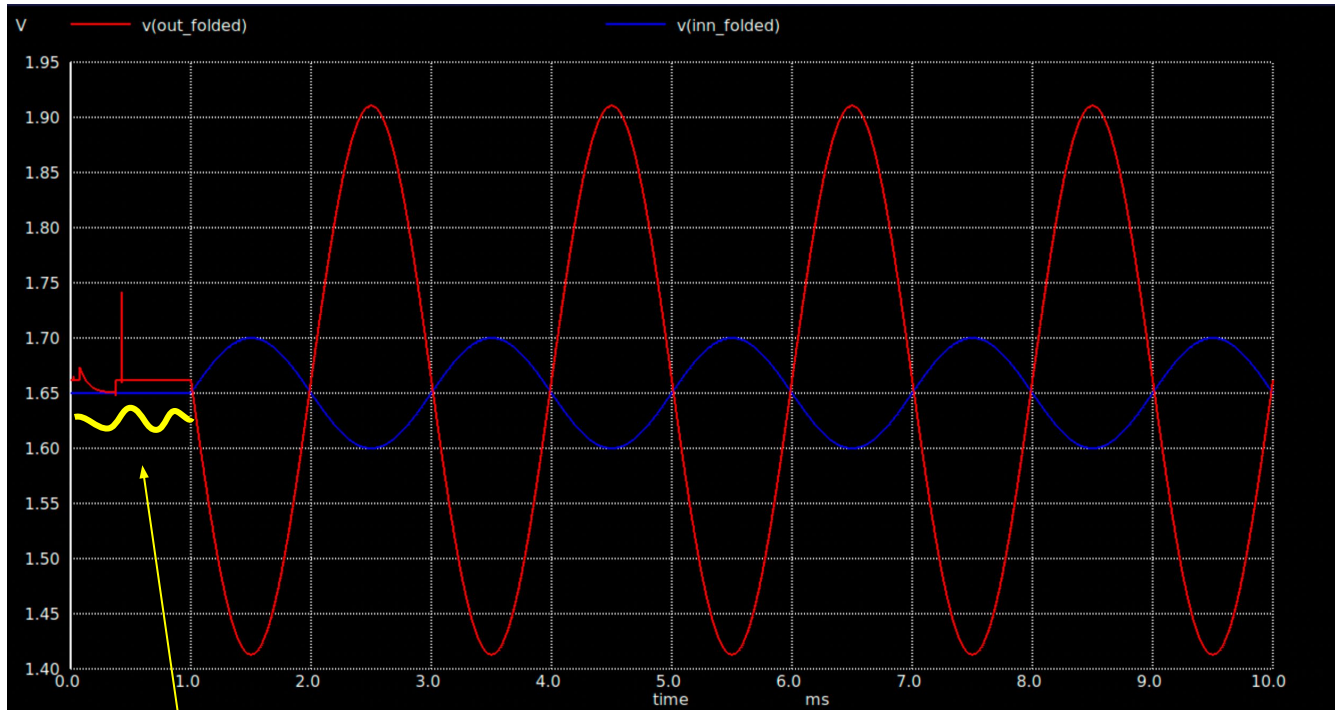


VDD, GND, SCAN INPUT

Scan chain input artificially generated to enable folded cascode 3x sizing



System Simulation Results

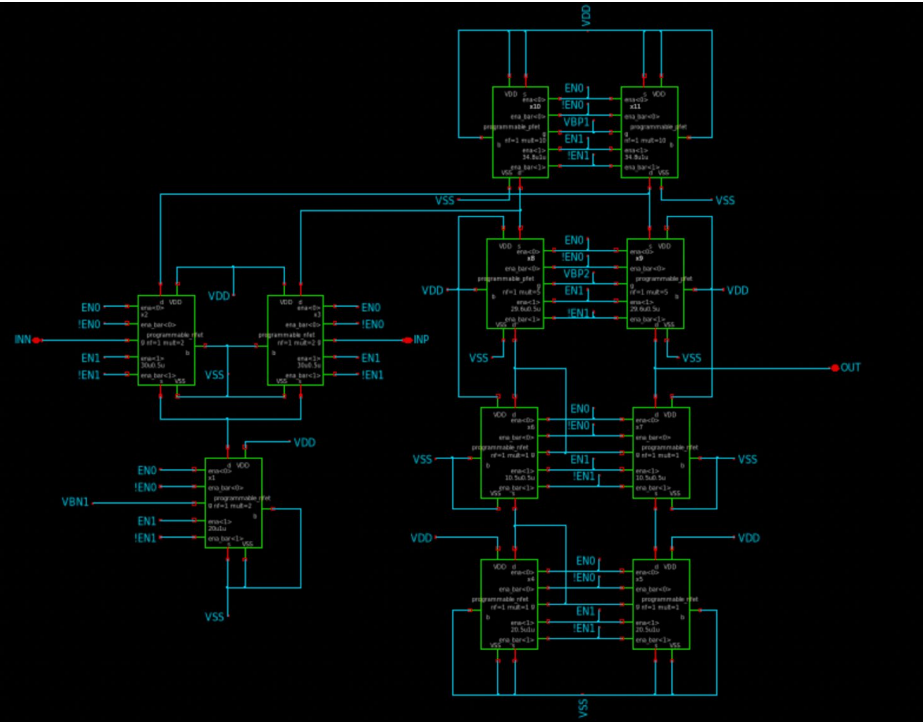
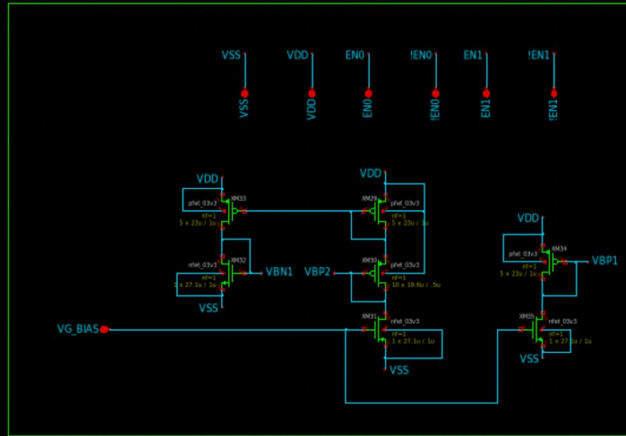


Measured Gain of
4.975 (Targeting 5)

Start Up (Scan In Propagating Through Scan Chain)

Programmable Folded Cascode Schematic

Bias Voltage Generation and Pins



Sizing and Device Information

Device	Role	W (μm)	L (μm)	ID (μA)
XM10/XM11	NMOS input pair	60	0.5	75
XM12	NMOS tail	40	1	150
XM2/XM5	PMOS current sources	174	1	150
XM3/XM4	PMOS cascodes	148	0.5	75
XM6/XM7	NMOS cascodes	10.5	0.5	75
XM8/XM9	NMOS mirrors	20.5	1	75

- VDD = 3.3 V
- IBIAS = 100 μA
- 3.3 V Devices
- Sizing for the original amplifier and base sizing for programmable amplifier is listed in the table
- Total current consumption is approximately 300 μA , giving roughly 1mW at 3.3V

AC Open-Loop

Amplifier Performance

-Driving Ideal 120 pF load capacitance
-Intended load of 100 pF off-chip,
10-20 pF added for parasitic modeling

1x Configuration:

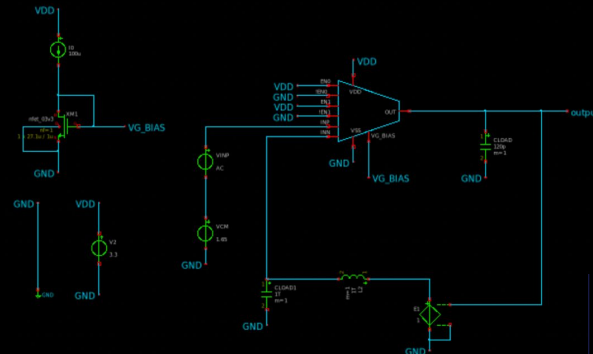
DC Gain: 76.0 dB
GBW: 1.47 MHz
Phase Margin: 86.7 Degrees

2x Configuration:

DC Gain: 76.0 dB
GBW: 2.80 MHz
Phase Margin: 83.5 Degrees

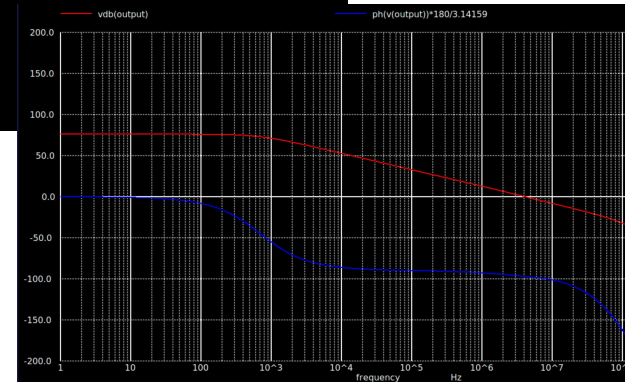
3x Configuration:

DC Gain: 76.0 dB
GBW: 4.15 MHz
Phase Margin: 83.0 Degrees



```
Models
include $:/180MCU_MODELS/design.ngspice
lib $:/180MCU_MODELS/sm141864.ngspice typical

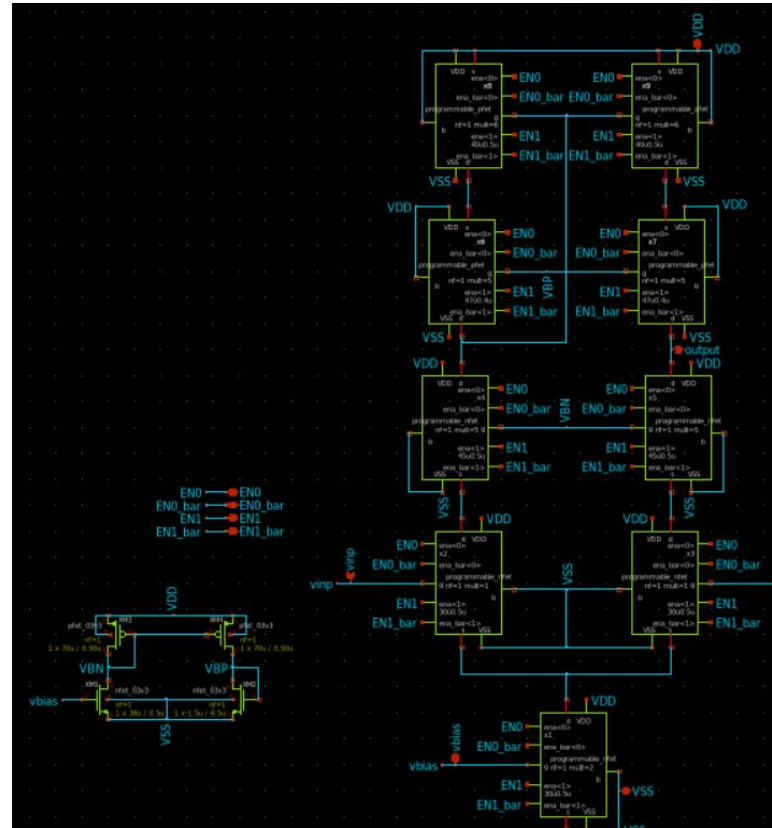
Simulation
.control
ac dec 100 1 1G
plot vdb(output) ph(v(output))*180/3.14159
meas ac dc gain find vdb(output) at=1
meas ac gbw when vdb(output)=0
let gain_db = vdb(output)
let phase = 180/3.14159*ph(v(output))
meas ac phase at gbw find phase when vdb(output)=0
let pm = 180 + phase_at_gb
print pm
.endc
```



Simulation
Result from 3x

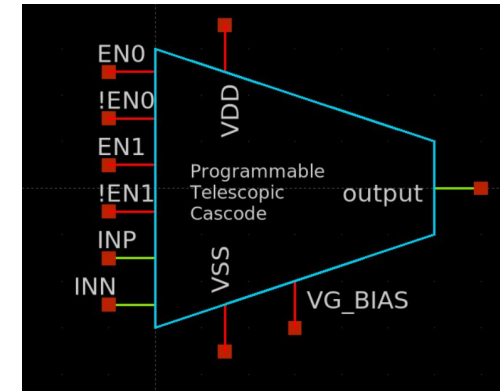


Programmable Telescopic Cascode



ID	Programmable MOSFET	W (μm) (1x)	L (μm) (1x)	Multiplier
X1	Programmable	30	.5	2
X2, X3	Programmable	30	.5	1
X4, X5	Programmable	30	.5	6
X6, X7	Programmable	47	.4	5
X8, X9	Programmable	30	.5	8
XM3, XM4	PMOS	70	.5	1
XM2	NMOS	1.5	.5	1
XM1	NMOS	30	.5	1

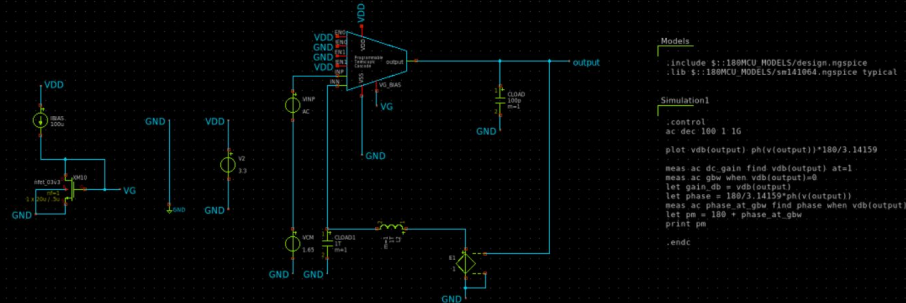
Symbol View



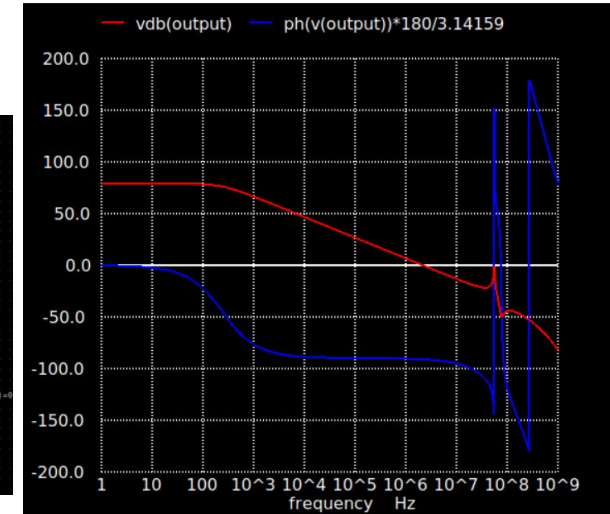
AC Open Loop

Amplifier Performance

- Amplifier driven by off chip current modeled by IBIAS, driving an off chip 100 pF capacitor
- 1x | DC Gain: 79.16 | GBW: 2.19 MHz | PM: 88.88
- 2x | DC Gain: 79.16 | GBW: 4.38 MHz | PM: 87.27
- 3x | DC Gain: 79.16 | GBW: 6.57 MHz | PM: 86.56
- GBW increase by 2x and 3x as expected by driving 2x and 3x current respectively

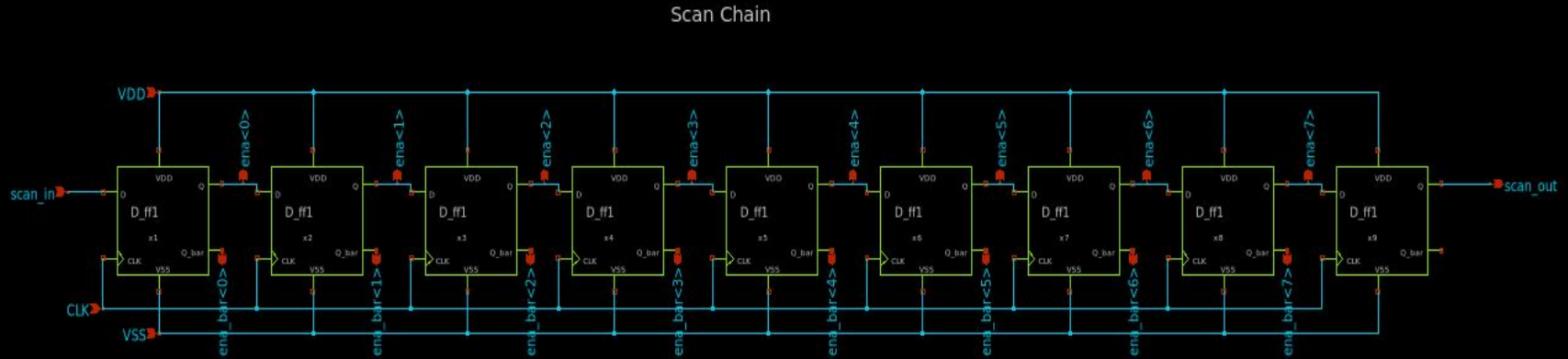


1X Gain and GBW match that of
non-programmable telescopic cascode
found in extended slide deck

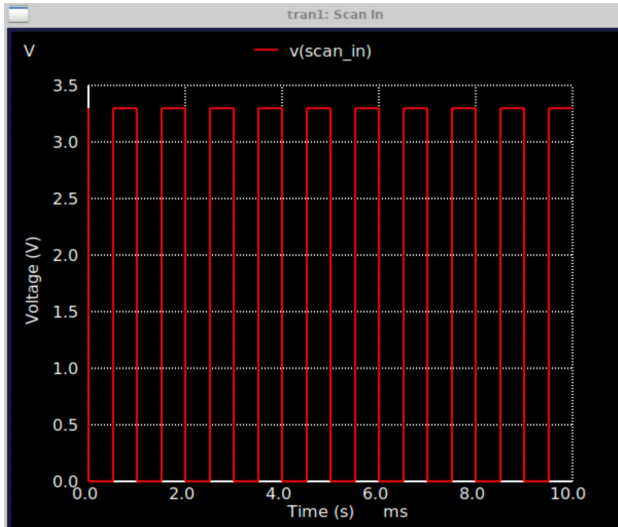


1X Bode Plot

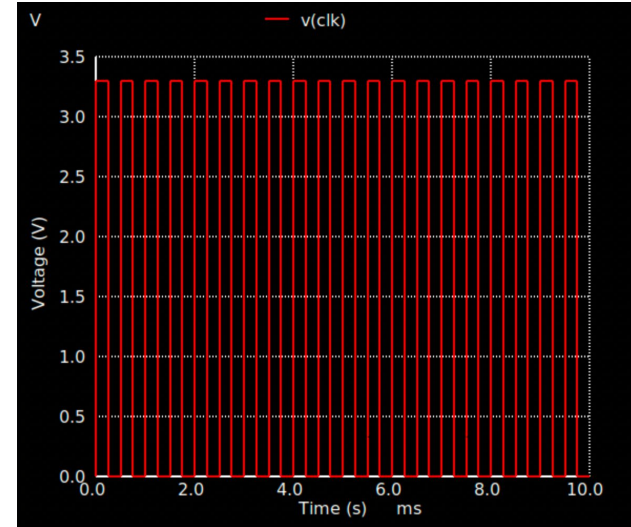
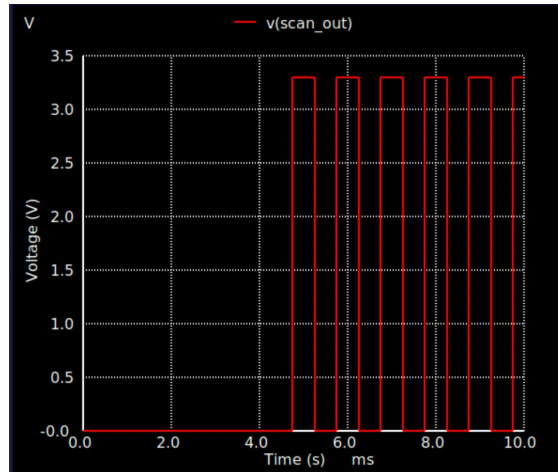
Scan Chain



Scan Chain Simulation



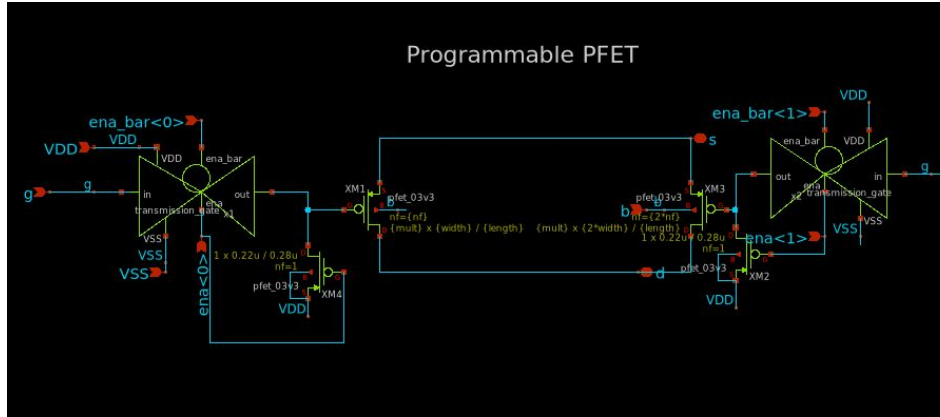
- Scan out does not match scan in until 9 clock cycles due to there being 9 flip flops



- Freq = 2kHz
- Has been tested and working at 20 kHz

Programmable Transistors

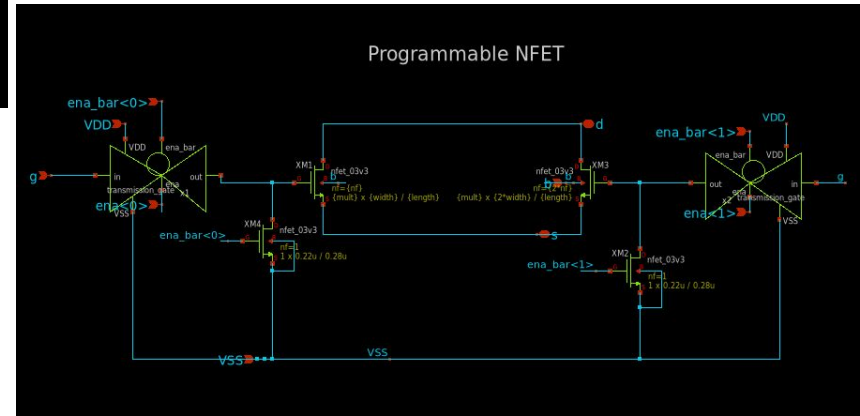
Programmable PFET



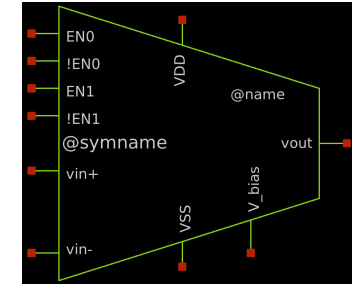
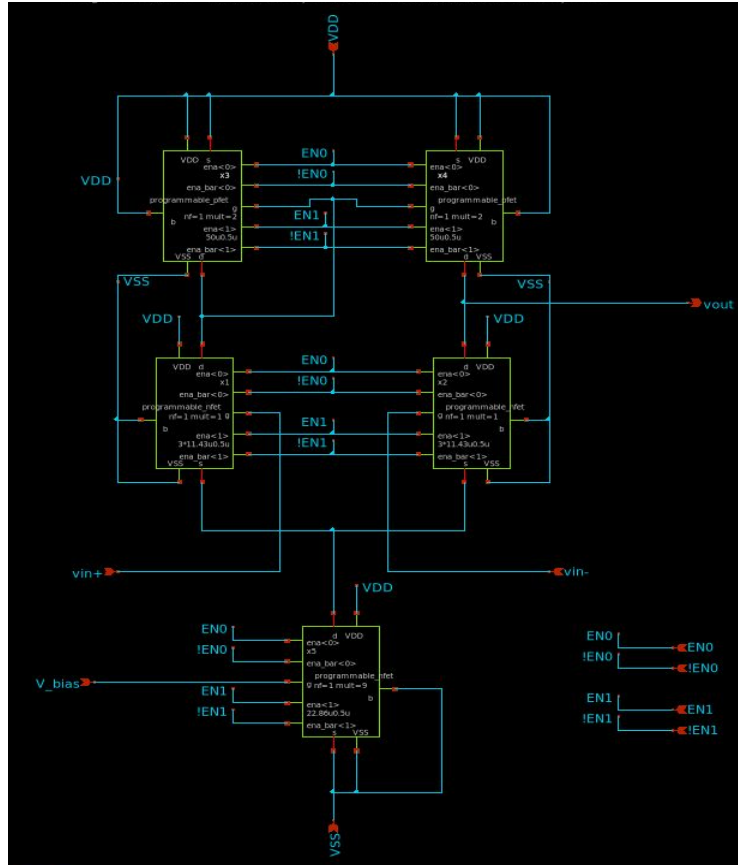
Pull Up transistors to avoid floating gates

Pull Down transistors to avoid floating gates

Programmable NFET

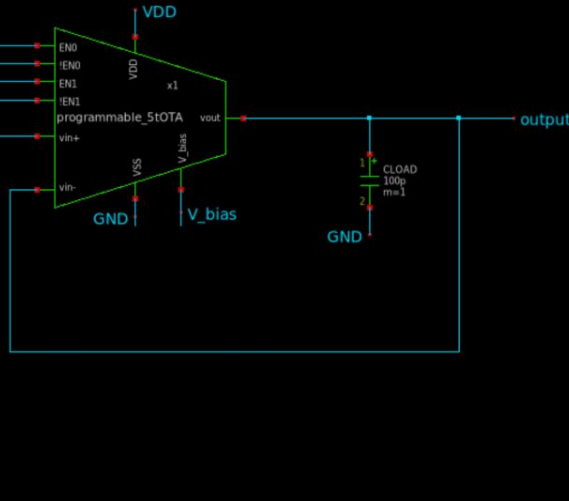
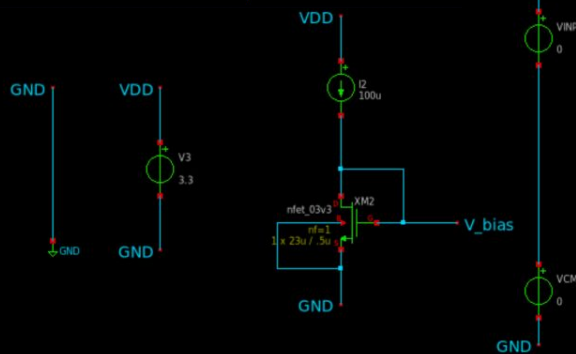
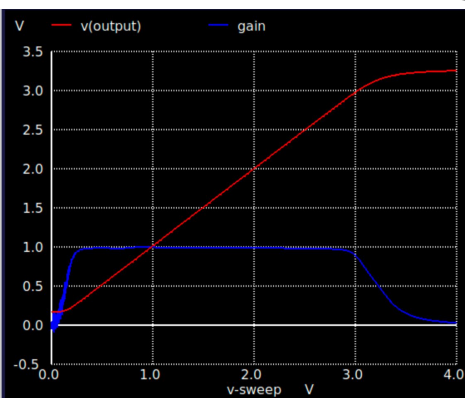


Programmable 5-Transistor OTA



ID	Programmable MOSFET	W (μm) (1x)	L (μm) (1x)	Multiplier
X3, X4	Programmable (PFET)	50	.5	2
X1, X2	Programmable (NFET)	11.43	.5	1
X5	Programmable (NFET)	22.86	.5	9

Programmable 5-Transistor OTA - DC Unity



Amplifier Performance(DC)

-Driving Ideal 100 pF load capacitance

1xW/L

ICMR: 0.298V - 2.924V (2.626V range)

Output Swing: 0.298V - 2.924V (2.626V range)

2xW/L

ICMR: 0.271V - 2.924V (2.653V range)

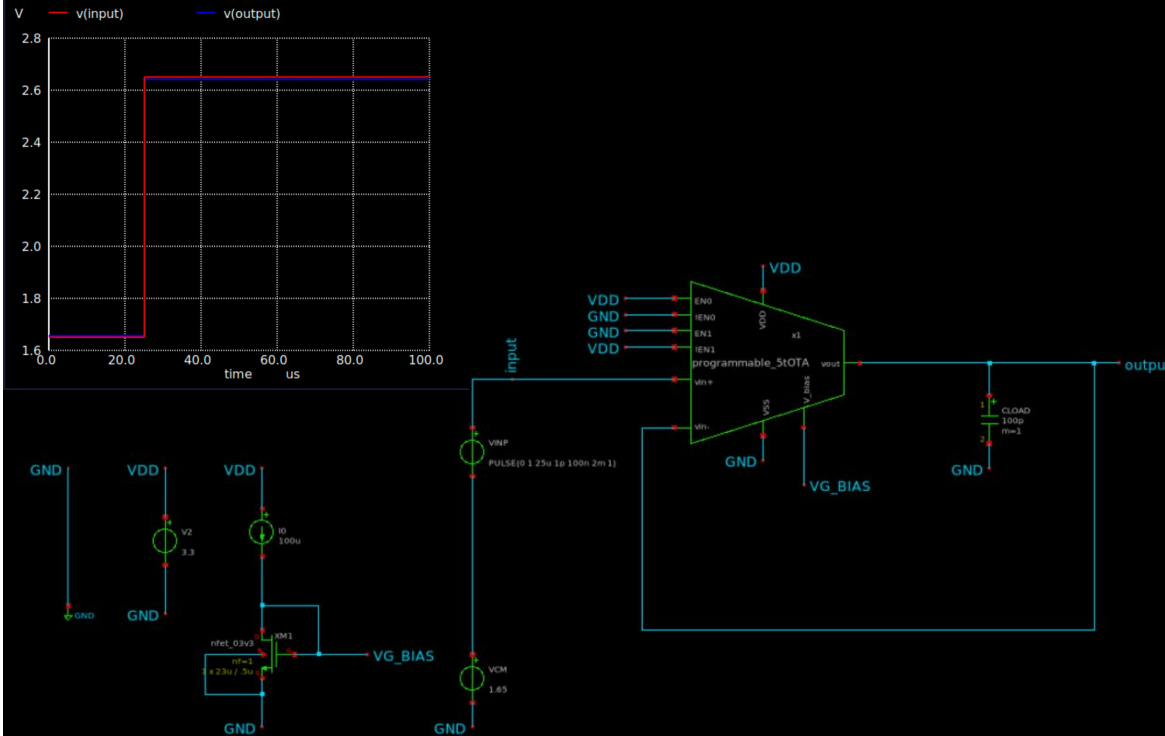
Output Swing: 0.271V - 2.924V (2.653V range)

3xW/L

ICMR: 0.261V - 2.924V (2.663V range)

Output Swing: 0.261V - 2.924V (2.663V range)

Programmable 5-Transistor OTA - Slew Rate



Amplifier Performance(Transient)

-Driving Ideal 100 pF load capacitance

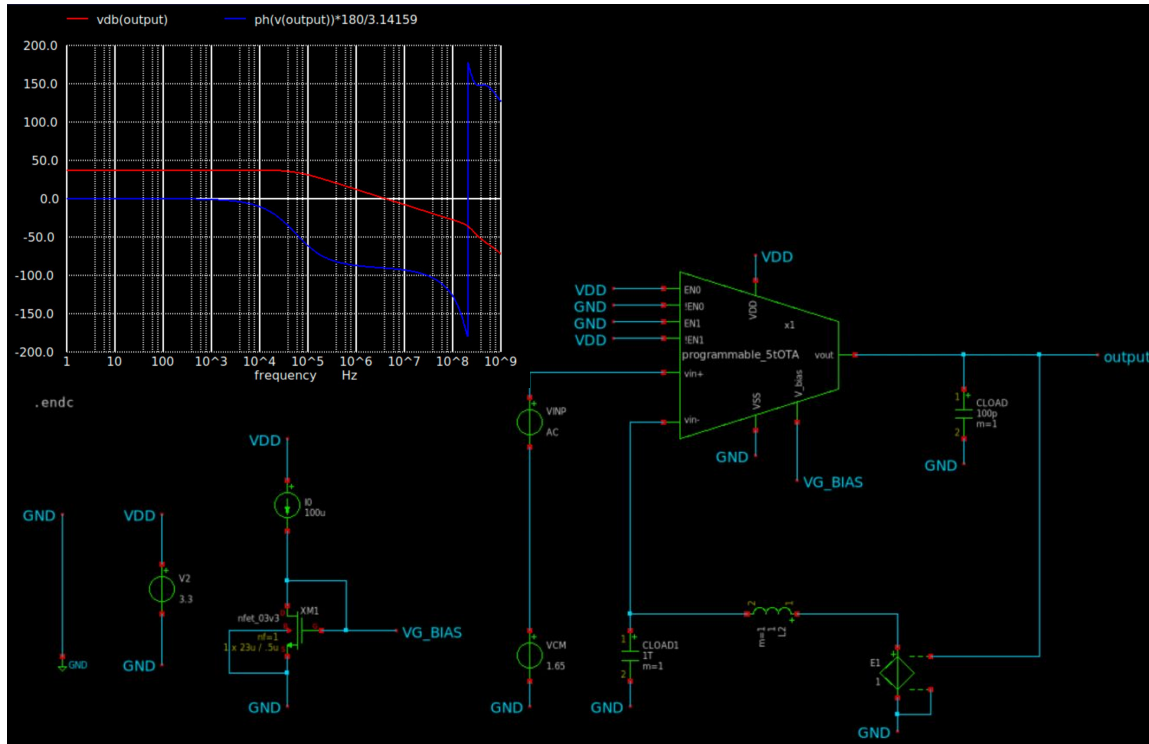
Unity Step Response (1V step, 1.65V $\rightarrow$ 2.65V):

1xW/L
Slew Rate: 8.15 V/ μ s

2xW/L
Slew Rate: 17.05 V/ μ s

3xW/L
Slew Rate: 26.70 V/ μ s

Programmable 5-Transistor OTA AC



Amplifier Performance(AC)

-Driving Ideal 100 pF load capacitance

1xW/L
 DC Gain: 37.5 dB
 3-dB BW: 54.8 kHz
 GBW: 4.12 MHz
 PM: ~90.0 Degrees

2xW/L
 DC Gain: 37.5 dB
 3-dB BW: 107.3 kHz
 GBW: 8.08 MHz
 PM: ~90.0 Degrees

3xW/L
 DC Gain: 37.5 dB
 3-dB BW: 159.8 kHz
 GBW: 12.0 MHz
 PM: ~90.0 Degrees

Next Steps

- Continue to Test System-Level Configurations and Specifications
- Experiment and Document Two-Stage Configurations, Compensation
- More Characterization Simulations (Noise and PSRR)
- Begin Layout

Thank You!

Any feedback is greatly appreciated. Please view our full slide deck linked on our repository and issue for more simulations and calculations.